



CI/CD on AWS

Deploy Everything All Of The Time

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Why does DevOps matter?

- High-performing IT organizations deploy **30x** more frequently with **200x** shorter lead times; they have **60x** fewer failures and recover **168x** faster.
- Lean management and continuous delivery practices create the conditions for delivering value faster, sustainably.
- High performance is achievable whether your apps are greenfield, brownfield, or legacy.

What are CI and CD?

- Continuous Delivery
 - A software engineering approach in which teams keep producing valuable software in short cycles and ensure that the software can be reliably released at any time
- Continuous Integration
 - The practice, in software engineering, of merging all developer working copies to a shared mainline several times a day

What are CI and CD?

- Changes to code automatically deployed to mainline branch
 - After passing unit and mock tests
- Makes changes to code and deployments iterative, not monolithic
- Bugs are detected quickly
- Helps automate deployments
- Allows rapid development and deployment

Immutable Infrastructure

SOURCE CODE
REPOSITORY



SUBMIT
CODE

CONTINUOUS
INTEGRATION SERVER



CLOUD FORMATION



CODE
REPOSITORIES
BUILDS & PACKAGES
BUILD OUTPUT



DEV FACING
NOTIFICATIONS
TEST
RESULTS



PROJECT MANAGEMENT
SERVER



DEVELOPER

SCHEDULE
BUILD

PICK
TASKS



Components of traditional CI/CD on AWS



Deployment
Software
On Laptop



Git Repository
EC2 Instance



CI Software
EC2 Instance



Project Management
Software
EC2 Instance

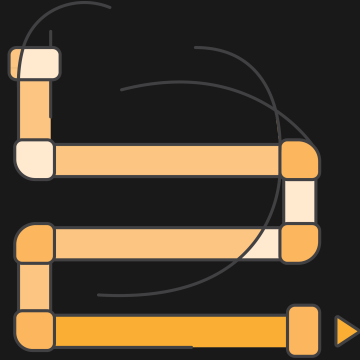


Build
Artifacts
S3 Bucket

AWS code services



AWS CodeCommit

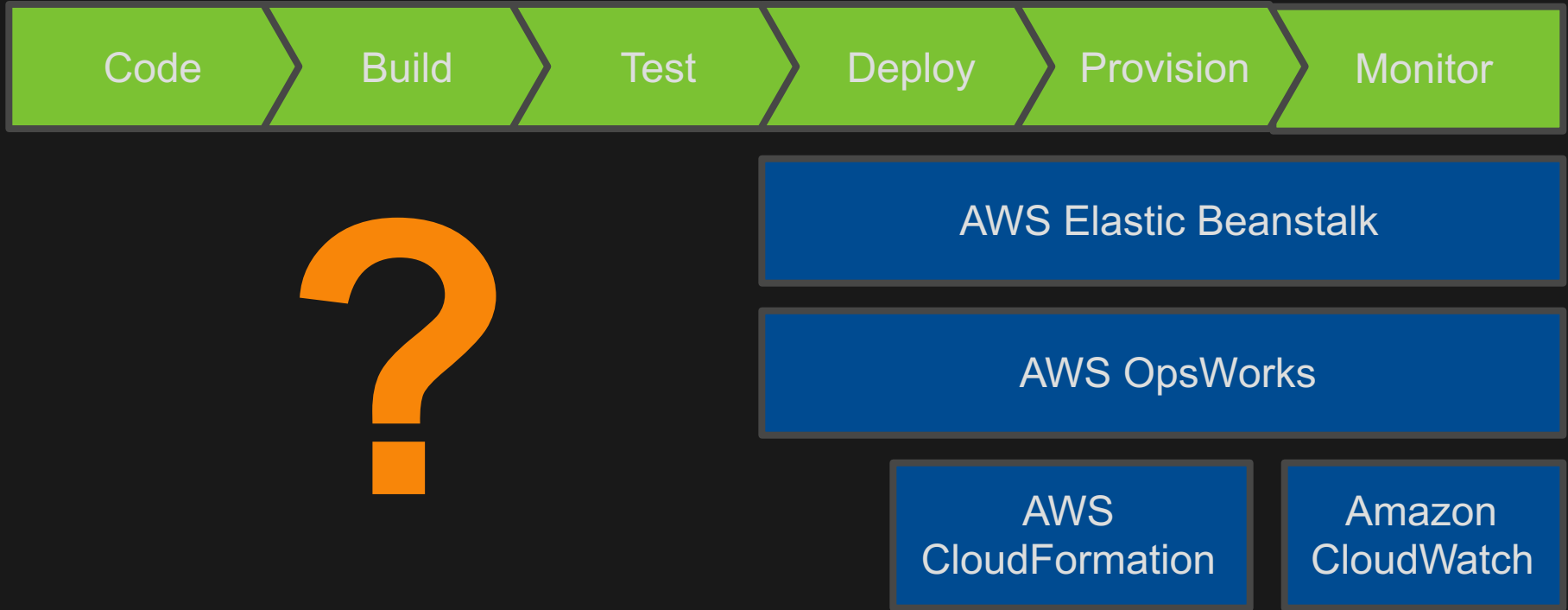


AWS CodePipeline



AWS CodeDeploy

Cloud software development lifecycle



CodeCommit: source control

Private Git repositories hosted on Amazon S3

Full Git compatibility (use with existing tools)

All the benefits of the cloud

(scalable, durable, reliable, low pay-as-you-go pricing)

No size limits on repositories (store binary files)

Online web console

CodeCommit: same Git experience

```
$ git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/aws-cli
Cloning into 'aws-cli'...
Receiving objects: 100% (16032/16032), 5.55 MiB | 1.25 MiB/s, done.
Resolving deltas: 100% (9900/9900), done.
Checking connectivity... done.
$ nano README.rst
$ git commit -am 'updated README'
[master 4fa0318] updated README
 1 file changed, 1 insertion(+)
$ git push
Counting objects: 3, done.
Delta compression using up to 4 threads.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 297 bytes | 0 bytes/s, done.
Total 3 (delta 2), reused 0 (delta 0)
remote:
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/aws-cli
 4dacd6d..4fa0318  master -> master
```

CodePipeline: continuous delivery

Customizable release automation, with integrated build and test

Model and visualize custom release workflow
(source → build → beta → gamma → prod)

Automate builds, tests, and deployments

Integrate with third-party and custom tools

Deployment

Coordinate software updates to fleets of EC2 instances

Rolling updates for no downtime

Deployment health checks and easy rollback

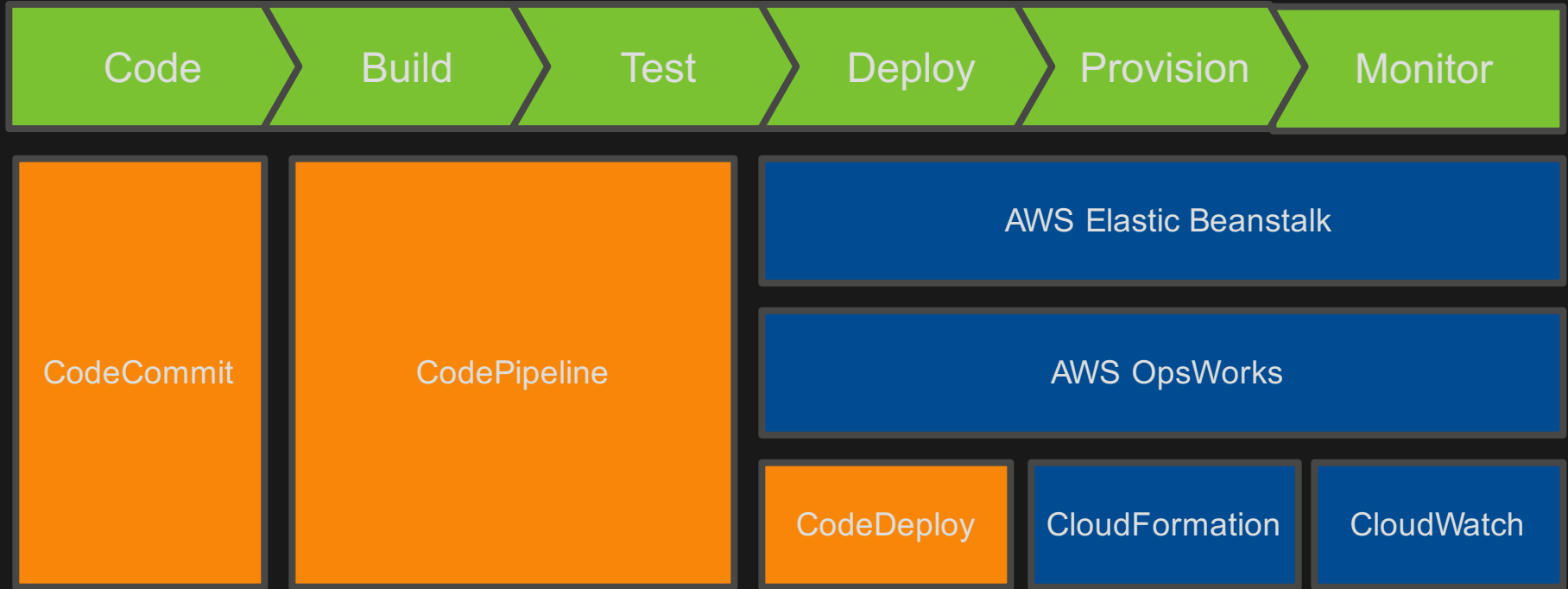
Auto Scaling integration

Works with any application

Reuse existing setup tools

(Bash, Powershell, Chef, Puppet...)

Cloud software development lifecycle



Components of CI/CD with AWS Code Services



**AWS
CodePipeline**



**Git Repository
CodeCommit**



**AWS
CodeDeploy**



**CI Software*
EC2 Instance**



**Project Management
Software
EC2 Instance**



**Build
Artifacts
S3 Bucket**

DevOps Partner Solutions

Solutions vetted by the AWS Partner Competency Program



Continuous Integration & Delivery

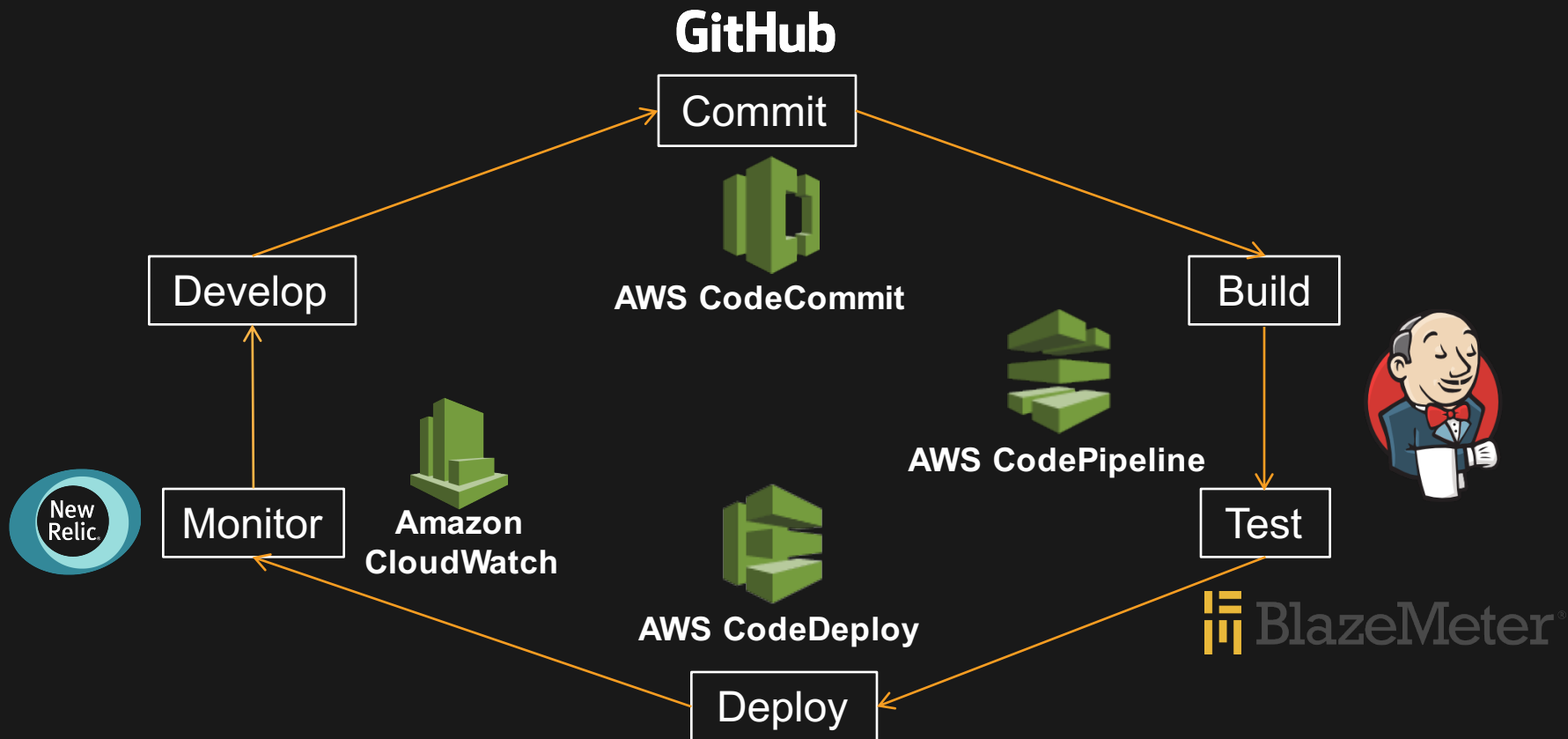
Rapidly and reliably build, test and release your applications,
while improving quality and reducing time to market

Configuration Management

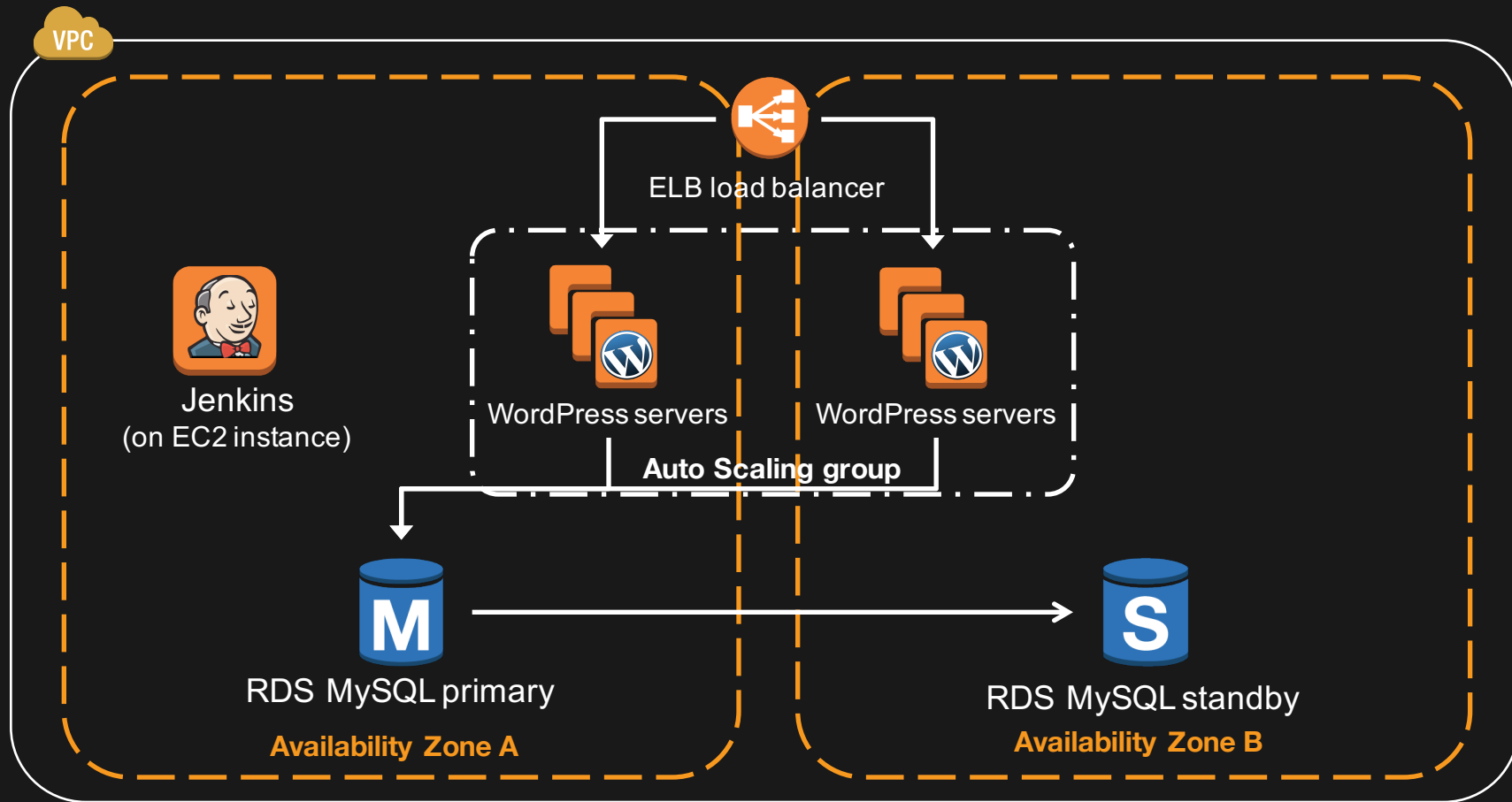
Treat your infrastructure as code, manage container,
operating system, and application configurations



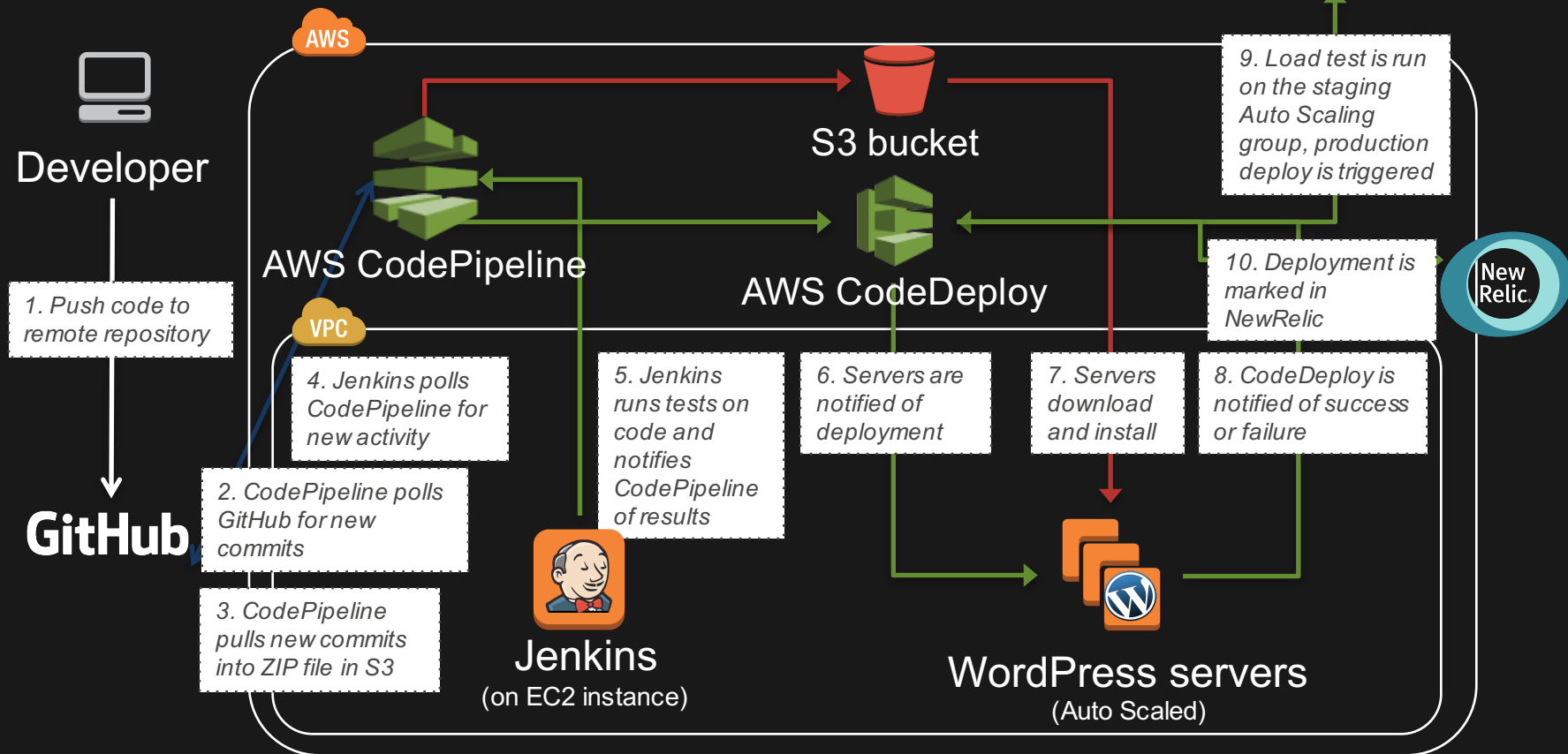
Today's demo



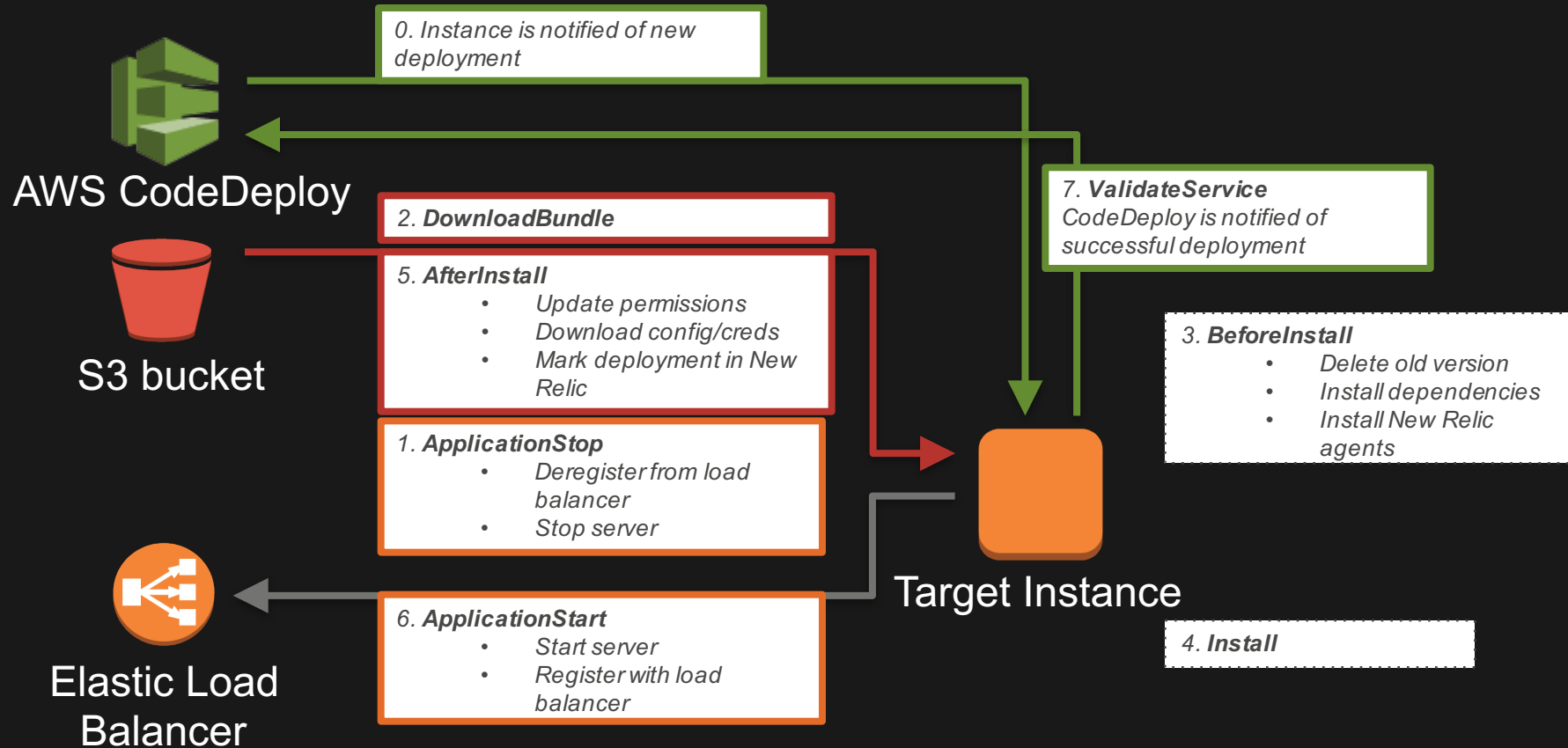
Application architecture



Deployment overview

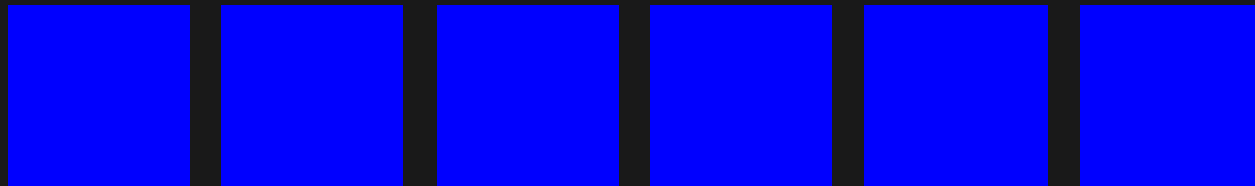


CodeDeploy lifecycle events

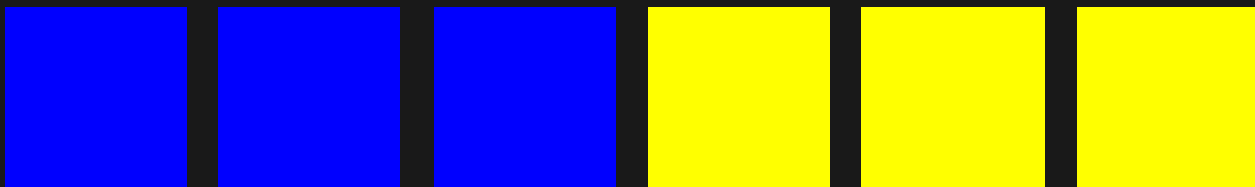


CodeDeploy default deployment configs

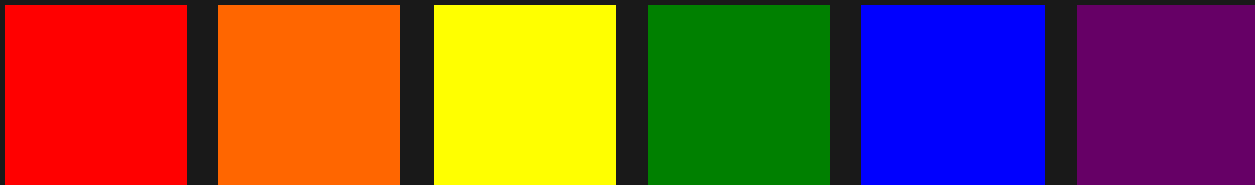
AllAtOnce



HalfAtATime



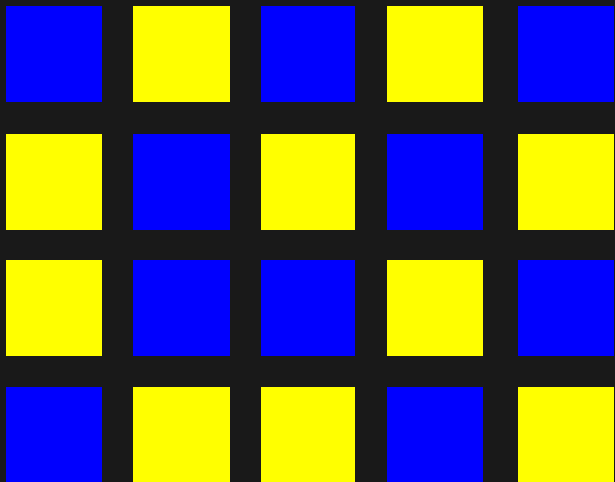
OneAtATime



CodeDeploy custom deployment configs

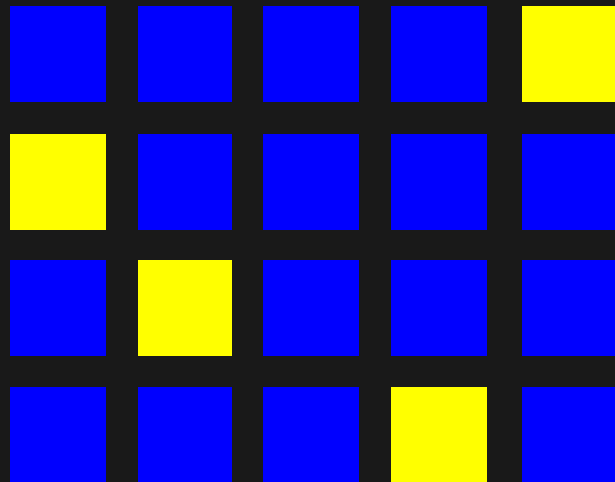
HOST_COUNT = 10

No less than 10 hosts should be healthy during deployment,
regardless of the number of instances

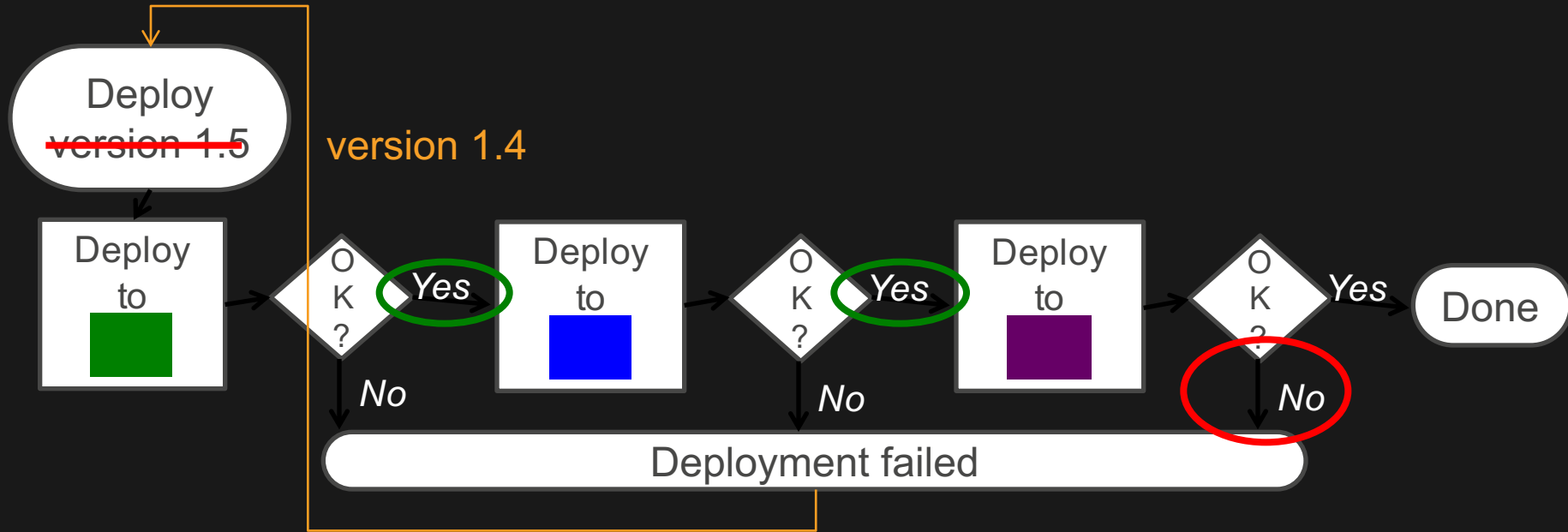


FLEET_PERCENT = 80

No less than 80% of hosts should be healthy during deployment



Failed deployments in CodeDeploy



Recover from a failed or canceled deployment by creating a new deployment of the previous version

Live demo

